Predicting Olympic Medal Success: Paris 2024

IS 507

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Motivation: Why This Problem?

- Olympic medals represent exceptional athletic achievement, but predicting winners is difficult.
- Medal success depends on several factors, including athlete characteristics, sport, and national representation.
- Machine learning provides an opportunity to examine patterns associated with medal success.
- This project also explores the challenge of predicting relatively rare outcomes.

Research Question

“ Can we predict whether an athlete will win an Olympic medal based on their characteristics and event information? ”

Potential impact

- Identify factors associated with medal-winning performance.
- Help sports analysts understand patterns in athletic success.
- Support data-informed decisions by sports organizations.
- Provide a foundation for predicting outcomes at future Olympic Games.

Formal Prediction Problem: X, Y, and Loss

Component	Definition
Problem type	Binary classification
X (inputs)	Age, gender, height, weight, country, sport, team-event status
Y (target)	Medal won (1) or no medal (0)
Loss function	Binary cross-entropy / log loss
Evaluation	PR-AUC, precision, recall, F1-score

Dataset Source and Feasibility

Primary dataset: Paris 2024 Olympics Medals — Kaggle

- Focuses on medal results from the Paris 2024 Olympics.
- Provides a starting point for analyzing Olympic medal success.
- Additional athlete-level participation data may be needed.
- Exact number of rows, columns, and missing values must be verified.

What Does the Data Look Like?

Feature	Data type	Example
Age	Numerical	24
Gender	Categorical	Female
Height	Numerical	175 cm
Weight	Numerical	68 kg
Country	Categorical	USA
Discipline	Categorical	Swimming
Team event	Binary	0 or 1
Won medal	Binary target	0 or 1

The planned modeling dataset will contain one row per athlete.

Data Challenges

1. **Missing values:** Some athletes may have missing height, weight, or other information.
2. **Class imbalance:** Non-medalists are expected to outnumber medalists.
3. **Multiple events:** An athlete may compete in more than one event.
4. **Team events:** Multiple teammates may receive medals for the same result.
5. **Data leakage:** Medal totals or post-event information could reveal the outcome.

Related Work and Research Gap

Existing work:

- Olympic datasets are widely used to study medal counts and historical trends.
- Athlete-level historical data allows researchers to examine participation and performance.
- Classification models can be used to investigate factors associated with sporting outcomes.

Research gap:

- Medal counts alone do not explain individual athlete outcomes.
- A single Olympic Games dataset lacks historical performance information.
- This project will explore how well basic athlete characteristics predict medal success.

Additional data source: 120 Years of Olympic History — Kaggle

Baseline Models

Baseline 1: Majority-Class Classifier

- Predicts "No Medal" for every athlete.
- Establishes a simple reference point.
- Demonstrates why accuracy alone can be misleading.

Baseline 2: Logistic Regression

- Uses athlete and event features.
- Estimates the probability of winning a medal.
- Provides an interpretable benchmark for complex models.

Model Training and Evaluation Plan

1. **DATA PREPARATION**

Clean, merge, handle missing values

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2. **FEATURE ENGINEERING**

Encode categories, scale numeric features

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3. **TRAIN / VALIDATION / TEST**

Prevent leakage and preserve class balance

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4. **MODEL TRAINING**

Dummy classifier, logistic regression, advanced models

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5. **EVALUATION**

PR-AUC, precision, recall, F1-score

Models and Planned Improvements

Model	Why consider it?
Logistic Regression	Simple and interpretable baseline
Random Forest	Captures nonlinear relationships
XGBoost / LightGBM	Models complex feature interactions
Class-weighted models	Address class imbalance

Planned Improvements:

- Engineer new features, such as age at the Games and discipline-relative physical measurements.
- Tune hyperparameters using cross-validation.
- Optimize classification thresholds using validation data.
- Compare models to determine whether complexity improves performance.

Preliminary Progress and Next Steps

Current stage

- Research question and prediction target defined.
- Initial dataset identified.
- Baseline models selected.
- Data challenges and evaluation approach outlined.

Next steps

- Verify dataset size, variables, and completeness.
- Obtain non-medalist records if necessary.
- Prepare and merge the data.
- Implement and evaluate baseline models.
- Compare results and analyze limitations.

References

1. Berkay Alan. Paris 2024 Olympics Medals Dataset. Kaggle.

2. 120 Years of Olympic History: Athletes and Results. Kaggle.

3. Predicting Olympic Medal Success: Paris 2024. Preliminary Project Report, October 2026.

Thank You